

DC Circuits

Nathan Warner



Northern Illinois
University

Computer Science
Northern Illinois University
United States

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Basic quantities

- **Planetary model of the atom:** One of the major issues with the planetary model is the idea that electrons whirl around the nucleus in stable, planet-like orbits. That's simply not true. First, the electron inhabits a region of 3D space and does not simply move through a plane

Second, due to the Heisenberg Uncertainty Principle, we can't precisely plot the position and trajectory of a given electron. The best we can do is make a plot of where the electron is likely to be. This is called a probability contour

To understand this concept, imagine that you could record the position of a specific electron relative to the nucleus at a specific time. A moment later you record its new position, a moment after that you record the next position, and so on for thousands of measurements. If you attempted to plot them all, you would wind up with a cloud of dots around the nucleus. This cloud is referred to as an orbital. You wouldn't know how the electron got from one position to the next but you would get a general idea of where it was likely to be. Ultimately, an orbital looks nothing like a planetary orbit.

- **Shells, subshells, and orbitals:** The electron cloud is referred to as an **orbital**.

We note that there are several potential orbitals. Due to quantum physics, only certain orbitals are allowed. The permissible electron energy levels are first grouped into shells, then subshells and finally orbitals. It is important to remember that orbitals indicate the electron energy level. That is, a higher orbital implies a higher energy level. Further, orbitals fill in first from lowest energy level to highest energy level.

- **Charge and current:** Charge is an attractive force denoted by the letter Q and has units of coulombs.

Electrons are negatively charged and protons are positively charged. All electrons and protons exhibit the same magnitude of charge

Further, opposite charges attract while like charges repel, similar to the poles of a magnet.

- **Moving charge:** It is possible to move charge from one point to another. The rate of charge movement over time is called current. It is denoted by the letter I and has units of amps

$$1 \text{ amp} = 1 \text{ coulomb}/1 \text{ second}.$$

Thus, current for current I , charge Q , and time t ,

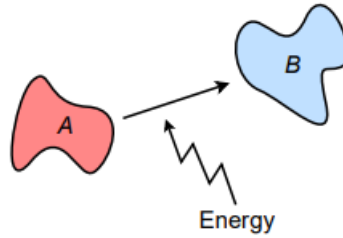
$$I = \frac{Q}{t}.$$

- **Energy and Voltage:** Energy is defined as the ability to do work. It is denoted by the letter W . The basic unit is the joule

If we were to move a charge from one point to another (for example, separating an electron from an atom), we would have to expend energy to do so

Figure 2.8

Defining voltage as work to move charge.



We would say that B has a higher electric potential than A. In other words, there is a potential difference between B and A. We refer to this change as voltage, denoted V , with units volts. One volt is one joule per coulomb.

$$V = \frac{W}{Q}.$$

- V is the voltage in volts,
- W is the energy in joules,
- Q is the charge in coulombs

As you might guess, the bigger the charge to be moved, the greater the energy required

- **Power and efficiency:** Power is the rate of energy usage

$$P = \frac{W}{t}.$$

Measured in watts.

$$1 \text{ watt} = 1 \text{ joule} / 1 \text{ second}.$$

- **The power law:** Recall that

$$I = \frac{Q}{t}, \quad V = \frac{W}{Q}, \quad \text{and} \quad P = \frac{W}{t}.$$

So,

$$P = \frac{VQ}{t} = IV.$$

We call $P = VI$ the **power law**.

- **Efficiency:** Efficiency is the ratio of useful output power to applied power expressed as a percentage

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\%.$$

If a device draws 200 watts of power and has a useful output of 120 watts, the efficiency is

$$\eta = \frac{120}{200} \cdot 100\% = 60\%.$$

In this case, the device is wasting 40% of the input power, or 80 watts.

- **amp-hour:** An amp hour (Ah, or ampere-hour) is a unit of measurement used to describe the total electric charge or capacity of a battery.

One amp hour means a battery can deliver a steady current of 1 amp for 1 hour.

The lifespan of a battery is roughly the amp-hour rating over the current draw.

$$\text{Lifespan} \approx \frac{Ah}{I}.$$

Amp-hour is calculated by multiplying current by time

$$Ah = It.$$

It measures the total electric charge a battery can hold or deliver.

- **Resistance and conductance:** Resistance is a measure of how difficult it is to establish a flow of current

Sometimes it is more convenient to use the inverse view of this phenomenon, and instead of referring to how difficult it is to establish a current, we would be interested in how easy it is to establish a current. This is called conductance and it is the reciprocal of resistance. Conductance is denoted by the letter G and has units of siemens (S).

$$R = \frac{1}{G}, \quad G = \frac{1}{R}.$$

- **Resistivity:** The measure of a material's inherent and general ability to restrict current flow is called resistivity. Resistivity is denoted by the Greek letter ρ

All other factors being equal, the higher the resistivity, the greater will be the resistance. Further, the greater the length of the material that the current must pass through, the greater the resistance. Finally, as the surface area of the cross section (i.e. the face) grows, the resistance decreases.

$$R = \frac{\rho \cdot l}{A}.$$

- R is the resistance in ohms,
- ρ is the resistivity,
- l is the length of the material,
- A is area of the face (h times w).

Resistivity is often specified in ohm-centimeters with the length and area similarly specified in centimeters and square centimeters

- **Resistors:** Resistors are **linear bilateral devices**. Being linear, their current-voltage relation can be drawn as a straight line. Bilateral means that the polarity of voltage or direction of current will not matter. In other words, unlike a battery, these devices cannot be inserted into a circuit backwards because either orientation works the same way. If the horizontal axis is voltage and the vertical is current, then the slope of the line yields the conductance

If the x-axis is labeled voltage, and y-axis is labeled current, the steeper the line, the lower its resistance

General purpose resistors use a color code to denote their value and tolerance. Typically, this will involve four color stripes: two for the precision/mantissa, one for the power of ten, and the fourth to indicate the tolerance. For higher precision, a five stripe version may be used with the first three denoting the precision/ mantissa. Alternately, high precision resistors may have their nominal value printed directly on them

Example of basic four stripe resistor color code.



Color	Numeral
	Black 0
	Brown 1
	Red 2
	Orange 3
	Yellow 4
	Green 5
	Blue 6
	Violet 7
	Gray 8
	White 9
<i>For small resistor values</i> If the multiplier band is gold then divide by 10; if silver, divide by 100.	

Note that the fourth band is spaced away from the others to avoid accidentally reversing the order.

- **Overvoltage (overload voltage):** Overvoltage is a condition where the electrical supply voltage rises above the safe design limit or rated nominal value of a circuit or device
- **Force Sensing Resistor (FSR):** The force sensing resistor consists of two layers of material with a nominal separation distance. It is presented as a flat membrane that is usually round or rectangular, and perhaps with an adhesive backing.

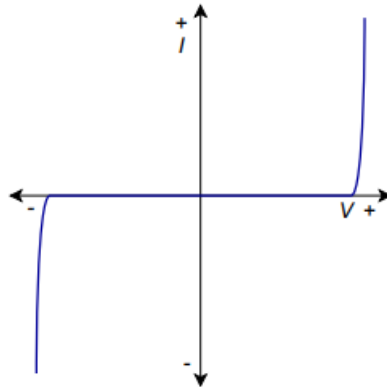
With no force applied, the device shows an extremely high resistance, well into the megohms. As force is applied to the surface, the two layers come into better contact which decreases the net resistance

- **Photoresistor (light dependent resistor (LDR)):** As their name implies, photoresistors are sensitive to changes in light level. They are also called LDRs, short for Light Dependent Resistor

In total darkness the device exhibits a very high resistance. As light levels increase, the resistance decreases.

- **Thermistor:** A thermistor is a device whose resistance is a function of temperature. These devices are available in two basic types. Either PTC, for Positive Temperature Coefficient; or NTC, for Negative Temperature Coefficient. PTC devices show an increase in resistance as temperature increases and NTC devices show a decrease in resistance as temperatures rise. Ideally, these are linear relationships with the plots showing straight lines.
- **Varistor:** Varistors are used as limiting devices, primarily to suppress unwanted voltage spikes in electronic equipment. The varistor is a unique device in that it has a highly nonlinear current-voltage characteristic.

Remember, when plotted with the voltage on the horizontal axis, the slope of the line represents the conductance. Consequently, the varistor shows a region of near zero conductance or extremely high resistance (the horizontal section), and two sections that are nearly vertical, which indicate extremely high conductance or near zero resistance. This characteristic allows the varistor to act as a limiting device.

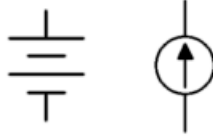


- **Schematic symbols:** Clearly, it would not be practical to create circuit drawings (schematics) using pictures of the actual devices. Instead, we use simple schematic symbols to represent them. There are two widely used schemes: ANSI (American National Standards Institute) and IEC (International Electrotechnical Commission). For many electronic components the ANSI and IEC versions are the same, but there are notable differences. ANSI tends to dominate in North America (surprise!) while IEC tends to dominate elsewhere.

Figure 2.38 shows the symbols for DC current and voltage sources. The long bar at the top of the voltage source is the positive terminal while the shorter bar at the bottom denotes the negative terminal. For the current source, the arrow points in the direction of conventional current flow. If the source is adjustable, it will often be shown with a diagonal arrow drawn through it. In some schematics, a DC voltage source will simply be drawn as a “connection dot” or node with the voltage labeled. It is assumed that the other end of the source is tied back to the system common or ground.

Figure 2.38

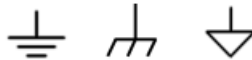
Voltage (left) and current source schematic symbols.



Speaking of ground, there are three different symbols used for a circuit's common connection point. These are: earth ground, chassis ground and signal or digital ground. These are shown in Figure 2.39. Earth ground is used when the circuit common is tied back to true earth (e.g., to the third pin on an AC receptacle). Chassis ground is a more generic term which would represent a common reference point that did not go back to true earth (e.g., a portable device). Finally, digital or signal ground is used to distinguish between common points where they might be separated for noise or interference concerns (e.g., a sensitive analog signal common separate from a digital logic common).

Figure 2.39

*Schematic symbols for ground.
(L-R) earth, chassis, signal.*



The symbols for resistors and other resistive devices vary considerably between ANSI and IEC versions. ANSI uses a zig-zag line while IEC uses a simple rectangle

One final note about component values: As decimal points are easy to lose on photocopies or small display screens, engineering notation multipliers are sometimes placed in the position of a decimal point. For example, a 4.7 k Ω resistor may be listed as simply 4k7.

Series Resistive Circuits

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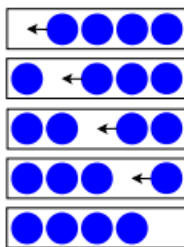
- **Intro:** Benjamin Franklin (pictured in Figure 3.1) began experimenting with the phenomenon of electricity in 1746. In 1752 he performed his famous kite experiment proving that lightning is a form of electricity by capturing charge from storm clouds in a leyden jar (an early form of an electrical charge storage device).

At this time the modern concept of an atomic model with electrons and protons did not exist and electricity was conceived of as a sort of fluid. Franklin surmised that the “electrical flow” moved from positive to negative. This idea was accepted and became the conventional view. Today we call this idea **conventional current flow**. In this model, current flows from a more positive voltage to a less positive voltage. We know now that the electron is the charge carrier in metals and the electrons travel in the reverse direction. Essentially, Franklin guessed wrong. Electrons move from a lower potential to a higher potential. We call this model **electron flow**. For most work, engineers and technicians use conventional flow, although in some cases, such as the explanation of semiconductors, electron flow is easier to visualize for some people. In short, conventional flow exists for historical reasons, and it is the model used for most analyses, including this text.

You might think the current direction would make a big difference in an analysis; after all, it certainly makes a big difference if you drive a car in the wrong direction. It turns out that both forms will achieve the desired results, we just have to be consistent with the usage.

To better understand this, consider that the movement of a net negative charge in one direction can be thought of as a movement of a net positive charge in the other direction. That is, the movement of an electron creates a “hole” where it used to be and that hole is net positive.

Figure 3.2
Electron versus hole flow.



Here we start at the top with a tube of identical marbles, all pushed to the right. In each step below it we move a marble to the left, mimicking the flow of electrons in a circuit. When we reach the bottom, each marble has been pushed left by one place. We can also arrive at the bottom drawing by simply taking the right-most marble in the top drawing and inserting it to the extreme left by jumping over the other three marbles. Here's the important bit: instead of imagining the marbles moving left, we

can also think in terms of “negative space” and imagine the empty slot moving to the right. That’s hole flow. The two views are functionally identical as they lead to the same outcome.

- **Circuits:** The word circuit comes from the Latin root *circ*, meaning “ring” or “around”. An electrical circuit consists of at least one ring or loop through which current flows. For example, if we have a battery attached to a lamp the current exits the battery, flows through the lamp, and then returns to the other side of the battery creating a loop or completed circuit. Without a path back to the battery, current will not flow. Thus, if we cut one of the wires connecting the battery and lamp, there is no path for current and no current flows. This is referred to as an open circuit and is a common fault that occurs when electronic systems are dropped or struck forcefully. Obviously, this will tend to render the circuit unusable. In this example, the lamp will not illuminate.

The opposite of the open circuit is the short circuit. In a short circuit, an unintended alternate path for current flow exists and this also can create a malfunction. In the case of our battery and lamp, a short circuit can occur if a piece of wire or metal accidentally fell across the terminals of the lamp. The current would then have a high conductance (i.e., low resistance) path around the lamp. The vast majority of the current would take this low resistance path instead of the higher resistance path presented by the lamp. The result would be that the lamp would not light. In either the open or short case, the light does not function but there is an important difference: for the short circuit, excessive current will flow out of the battery because there is little to resist the flow of current. Thus, the battery will be drained very quickly. For the open, no current flows and thus the battery is not drained.

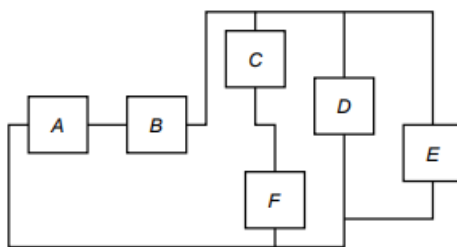
- **Series connections:** In general, a series connection is any connection of components configured such that the current through each component must be the same

Inside each of the lettered boxes would be a component such as a resistor or a voltage source. Note that for each component, there is one entrance point and one exit point. No matter which box you pick, the current flowing through it must be the same as the current flowing into the next box or out of the preceding box, and it doesn’t matter if you follow this path in a clockwise or counterclockwise fashion. Thus, this entire configuration is a series connection. The idea that current is consistent throughout should be selfevident. After all, the only way the current through, say, item B could be different from that flowing through item C or D is if some of it somehow “disappeared” along the way by magic. It is important to remember that consistent current is the hallmark feature defining a series connection:

It is possible that only a portion of a circuit exhibits a series connection.

Consider the more complex diagram presented in Figure 3.5. Some of these items are in series and some are not. For example, items A and B are in series with each other but not in series with the remaining items. Why? Because if we imagine the current flowing through A and then through B they must be the same, however, once beyond B, the current could split and flow down other branches: a portion entering C, a portion entering D and the remainder flowing into E. On the other hand, items C and F are in series with each other because whatever current is flowing through one of them must be flowing through the other. Thus, items A and B are in series with each other and items C and F are in series with each other, although all four are not in series as a group.

Figure 3.5
A more complex configuration.



It is possible that no two items in a circuit are strictly in series. Further, just because the currents through two items happen to be the same, that does not necessarily mean they are in series. Identical currents could be just a by-product of the component values chosen

Even if items D and E in Figure 3.5 have the same numeric value for current, we would not say that they are in series, anymore than we would say that any two people with the same last name would have to be siblings.

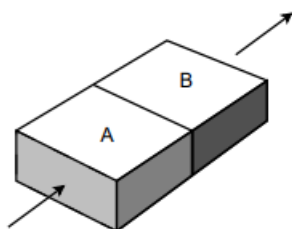
- **Combining Series Components:** Typically, a series connection will include multiple resistors.

Recall that resistors add

$$R = \frac{\rho \ell}{A}$$

resistance. If we consider two identical resistors placed in series, one after the other the effective length would double while keeping the resistivity and area unchanged. The combined result would be a doubling of the resistance of just one of them.

Figure 3.6
Resistors in series.



If we then generalize this to two arbitrary resistors of identical resistivity and area, the lengths would dictate the resistance of each, and the combined lengths would then

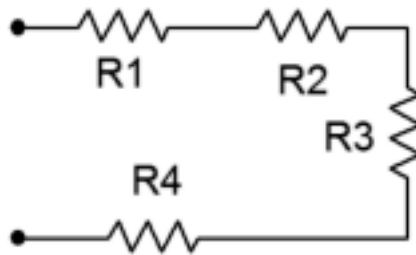
reflect the resistance of the pair.

$$R = \frac{\rho(\ell_1 + \cdots + \ell_n)}{A} = \frac{\rho}{A}(\ell_1 + \cdots + \ell_n)$$

Thus we find that the equivalent resistance of a group of series resistors is their sum:

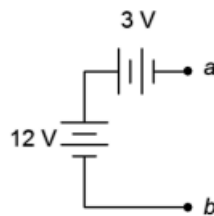
$$R_{\text{total}} = R_1 + R_2 + \cdots + R_n.$$

Consequently, as resistors in series add, total resistance may be found by summing the individual resistors.



Multiple voltage sources in series may also be added, however, polarities must be considered as opposing voltages partially cancel each other (i.e., adding a negative).

Figure 3.8
Voltage sources in series.



If we use point b as our reference, by inspection the top of the 12 volt source is 12 volts above point b (reminder, the long bar denotes the positive terminal). Also, by inspection, the right side of the 3 volt source (point a) is negative with respect to its left side. As the left side of this source is connected to the positive terminal of the 12 volt source, then it too must be 12 volts above point b. As its right side is 3 volts less than this side, point a must be 3 volts less than 12 volts, or 9 volts above point b. Thus

$$V_{ab} = 9\text{volts..}$$

Chasing this further, if the 3 volt source had been flipped so that it had the same polarity as the 12 volt source (positive toward the right) then $V_{ab} = 15$ volts. If the 12 volt source had been flipped (positive toward bottom) with the 3 volt source as drawn originally, then $V_{ab} = -15$ volts. If both sources had been flipped then $V_{ab} = -9$ volts. Finally, if point a had been taken as the reference then these four potentials would have the opposite polarity because, by definition, $V_{ba} = -V_{ab}$.

In contrast to voltage sources, differing current sources are not placed in series as they would each attempt to establish a different series current, a practical impossibility

- **Generality:**

$$\text{Effect} = \frac{\text{cause}}{\text{opposition}}.$$

- **Ohm's law:** We have

$$V = IR.$$

Or,

$$R = \frac{V}{I}.$$

Note that from this, we get

$$P = I^2 R = \frac{V^2}{R}.$$

- **Kirchoff's Voltage Law (KVL):** Along with Ohm's law, the key law governing series circuits is Kirchhoff's voltage law, or KVL. Named after nineteenth century German physicist Gustav Kirchhoff, this law states that the sum of voltage rises and voltage drops around a series loop must equal zero (the rises and drops having opposite polarities). Alternately, it may be reworded as the sum of voltage rises around a series loop must equal the sum of voltage drops

$$\sum V \uparrow = \sum V \downarrow.$$

- **Voltage divider rule (VDR):** An outgrowth of KVL is the voltage divider rule (VDR). In a series connection, the current is the same through each component. Thus, the voltage drops in a series connection must be directly proportional to the size of the resistances: the larger the resistor, the larger its voltage, and the larger its share of the total voltage applied to the series connection. Thus, the voltage across any resistor must equal the net supplied voltage times the ratio of the resistor of interest to the total resistance:

$$V_{R_x} = E \cdot \frac{R_x}{R_{\text{total}}}.$$

This is just two Ohm's law calculations:

$$V_{\text{total}} = E = IR_{\text{total}} \implies I = \frac{E}{R_{\text{total}}}.$$

Then, for a particular resistor R_x :

$$V_{R_x} = IR_x = \frac{E}{R_{\text{total}}} \cdot R_x = E \cdot \frac{R_x}{R_{\text{total}}}.$$

It is worth pointing out that "the resistor of interest" can, in fact, be the sum of multiple resistors in series.

While VDR is not required for any particular analysis, it serves two purposes: first, it saves some time because it skips over the computation of current, and second, it reinforces the ideal of a proportional division of voltage in a series connection. For example, if there are two resistors in series and one of them is twice the size of the other, then it must be the case that the larger resistor sees twice the voltage of the smaller resistor.

- **Series analysis:** Ohm's law, Kirchhoff's voltage law, the voltage divider rule and series component combinations are the tools we will use to solve general series circuit problems. There are multiple techniques for analyzing these circuits:
 - If all voltage source and resistor values are given, the circulating current can be found by dividing the equivalent voltage by the sum of the resistances. Once the current is found, Ohm's law can be used to find the voltage drops across individual resistors. Alternately, the individual voltage drops can be found using the voltage divider rule. At that point, power law can be used to find the power dissipation in each resistor or the power developed by the source(s).
 - If the circuit uses a current source instead of a voltage source, then the circulating current is known and the voltage drop across any resistor may be determined directly using Ohm's law.
 - If the problem concerns determining resistance values, the basic idea will be to use these rules in reverse. For example, if a resistor value is needed to set a specific current, the total required resistance can be determined from this current and the given voltage supply. The values of the other series resistors can then be subtracted from the total, yielding the required resistor value. Similarly, if the voltages across two resistors are known, as long as one of the resistance values is known, the other resistance can be determined using either the voltage divider rule or Ohm's law.
- **Node voltage:** A node voltage is the voltage at a given point with respect to some other point, normally ground

Note that as we treat connecting wires ideally (having no resistance), a node is the entire connection wire, not a specific location on it. Normally, simulators will automatically number each node in a circuit with ground being node 0. This method skips the entire business of inserting virtual meters in the circuit and reduces schematic clutter.

- **Voltage rise and voltage drop:** A voltage rise occurs when you move from a lower electric potential to a higher electric potential:

A voltage drop occurs when you move from a higher electric potential to a lower electric potential:

- **The Importance of Polarity:** A key item of importance when analyzing series circuits, or indeed any electrical circuit, is noting the polarity of the voltages.

Determining polarity for voltage sources is easy as their polarity is fixed (positive is always at the "long bar" and negative at the "short bar"). The polarity of any resistor will depend on the direction of current flow. To avoid confusion we will use the following standard:

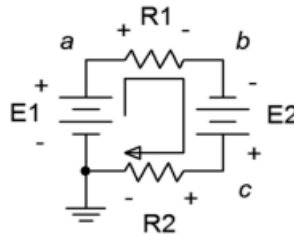
- Where conventional current enters a resistor, we label this point with a plus sign, and where the current leaves, a minus sign.

- During analysis, moving from plus to minus is a voltage drop. That is, we are moving from a more positive or higher potential to a less positive or lower potential, thus “dropping” voltage.
- Similarly, traversing from minus to plus is a voltage rise.

Remember, Kirchhoff’s voltage law states that the sum of voltage rises around a series loop must equal the sum of voltage drops.

Along with determining the voltage across single resistors, we are often interested in determining the voltage between arbitrary points in a circuit. These may span several components. It is imperative that we know the polarities of the individual voltages in order to determine the voltage between any two circuit points.

Figure 3.21
A multi-source, multi-resistor series circuit.



Here, the two voltage sources, E_1 and E_2 , aid each other because they are both trying to establish a clockwise current. This produces a net voltage of $E_1 + E_2$. Their polarities are fixed. The current direction will be as drawn because conventional current flows out of the positive terminal of a voltage source. The value of this current will be

$$I = \frac{E_1 + E_2}{R_1 + R_2}$$

via Ohm’s law.

Knowing the direction of current, the polarities of the voltage drops across the two resistors may be found. The point where the current enters the resistor is positive, and negative where it exits. Once these are labeled, either Ohm’s law or the voltage divider rule can be used to determine the voltages developed across the two resistors. Note that KVL states that the sum of these two resistor voltages must equal the net supplied voltage, or $E_1 + E_2$ in this case.

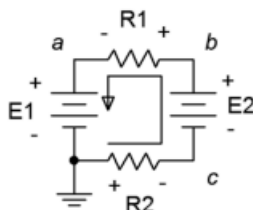
To determine the voltage from point a to point c , or V_{ac} , we start at point a and sum the voltage drops and rises until we get to point c . Here, the voltage across R_1 shows up as a drop

The result is the voltage across R_1 minus the voltage across E_2 .

Depending on the specific values, the sum may wind up being either a positive or negative value. If it's positive, this signifies that point a is at a higher potential than is point c . Conversely, if it's negative, this signifies that point a is at a lower potential than is point c .

To illustrate the importance of voltage source polarity, consider the series circuit shown in Figure 3.22.

Figure 3.22
Series circuit with altered polarity.



The two voltage sources oppose each other because their positive terminals are connected through R_1 , rather than one source's positive terminal aiding the other source.

Assume $E_2 > E_1$.

Following the indicated current direction around the loop:

- Passing through E_2 from c to b produces a voltage rise of $+E_2$.
- Passing through E_1 from a downward produces a voltage drop of $-E_1$.

Therefore, the net voltage driving the circuit is

$$E_{\text{net}} = E_2 - E_1.$$

The current is therefore

$$I = \frac{E_2 - E_1}{R_1 + R_2}.$$

Both resistors are in series, so the same current flows through them.

Because E_2 is the stronger source, conventional current leaves its positive terminal at b . It then travels:

$$b \rightarrow R_1 \rightarrow a \rightarrow E_1 \rightarrow R_2 \rightarrow c \rightarrow E_2.$$

Thus, current passes:

Through R_1 from right to left. Through E_1 from its positive terminal to its negative terminal. Through R_2 from left to right.

A source delivers energy when conventional current leaves its positive terminal. Current leaves the positive terminal of E_2 , so E_2 supplies power.

Current enters the positive terminal of E_1 , so E_1 absorbs power.

$$P_{E_1} = E_1 I.$$

If E_1 represents a rechargeable battery, that absorbed power corresponds to charging. If it is an ideal voltage source, “charging” simply means that it is absorbing energy.

- $E_1 < E_2$: current flows as drawn.
 - $E_1 > E_2$: current flows in the opposite direction.
 - $E_1 = E_2$: the ideal circuit has zero current.
- **Potentiometers and Rheostats:** Some applications require the use of variable resistances. One example is a basic volume control, but other examples include equipment calibration and user adjustable parameters such as the bass, treble and balance controls on a home audio system. Adjustable resistances fall under two categories, potentiometers and rheostats. Generally, these would be the same physical device, with the application and usage determining its name.

A potentiometer is a resistor with three leads. Along with the typical end leads, there is a third lead referred to as the wiper or tap. This third lead is connected to a small metallic arm that makes contact with the resistive material and which can be moved along the length of the material. Potentiometers, or pots for short, often are formed in a circular shape (rotary travel) but they can also be in the form of a straight line (linear travel)

The larger circular units with shafts and the straight unit at the top are designed to be adjusted by the end user (appropriately stylish knobs being attached to said shafts, of course), while the smaller units are intended to be adjusted only by a technician (note the small adjustment screw slots intended to receive fine screwdrivers). The smaller units are designed to be mounted directly on a printed circuit board while the user adjustable units are mounted on a front panel, and have wiring lugs instead of straight leads.

A pot can be visualized as two resistors in sequence with a tap between them. The total resistance never changes. What does change is the ratio of the two pieces.

$$R_{AB} = R_{AW} + R_{WB}.$$

Figure 3.30
Schematic symbol for a potentiometer, with labels.



- **Ganged potentiometers:** As a result, potentiometers are very useful as controllable voltage dividers (hence the name). Sometimes it is desirable to control two signals simultaneously from a single knob (e.g., one volume knob for a hi-fi stereo controls both the left and right channels). For this application, multiple resistive elements can be stacked and use a common shaft to control all of the wiper arms in tandem. These are known as **ganged potentiometers**.

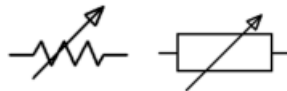
Finally, unlike fixed resistors, pots are available in limited sizes. Typical values use multiples of 1 or 5 for the total resistance

The output voltage at the wiper of a potentiometer changes because the wiper splits the total resistive track into two smaller resistor sections, and the voltage drops across each section in direct proportion to its length

- **Rheostat:** If only the wiper and only one of the two end terminals is used, then the device is referred to as a rheostat. Rheostats are often used to limit current. The value of the rheostat ranges from 0 up to the nominal value of the potentiometer.

Figure 3.31

Schematic symbols for a rheostat: ANSI (right) and IEC (left).

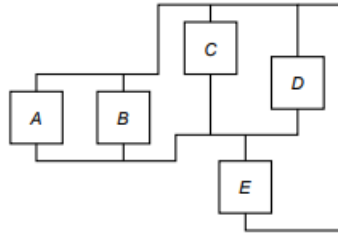


Parallel resistive circuits

- **The parallel connection:** In a parallel connection the voltage is the same across each component.

Figure 4.1

A parallel configuration.

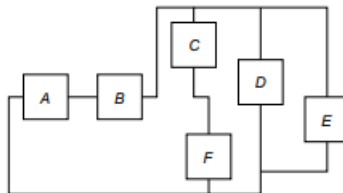


Although it may be drawn oddly, there are only two common connection nodes in this configuration: one along the top and the other along the bottom

If we were to take a voltmeter and place it across any of these blocks, the meter would read the same voltage across each because the probes are always contacting the same two nodes

Figure 4.2

A more complex configuration.



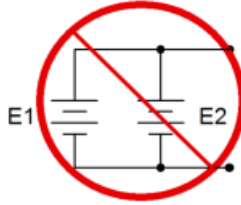
In this layout some elements are in series and some are in parallel. Only elements D and E are strictly in parallel here as they are the only two that connect to the same two nodes, and thus must see the same voltage.

Items C and F might at first appear to be in parallel with them but upon closer inspection it should be noted that C and F are in series with each other. That is, C and F will split the voltage that D or E sees, in accordance with the voltage divider rule. In other words, it is not true that the voltage across C or across F must be the same as the voltage across D or E

- **Combining Parallel Components:** Our first step is to determine how to combine parallel components in order to create a single equivalent component. Unlike series connections, this can be a little more time consuming.
- **Sources in parallel:** First, voltage sources are not placed in parallel as a general rule

Figure 4.3

Do not place voltage sources in parallel.

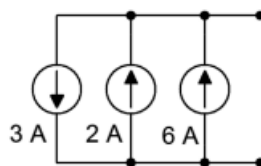


The reason is because a parallel connection requires the same voltage across each component. This would be impossible to achieve with each source trying to maintain a different voltage across the same two nodes. This may result in damage to the sources (e.g., exploding batteries). The main exception to this rule is if the sources have the same potential and the goal is to extend operational life (e.g., multiple D cell batteries in parallel).

- **Current sources in parallel:** When it comes to current sources, they may simply be added together, however, like series voltage sources, polarity is important. Referring to the three parallel current sources in Figure 4.4, the middle and right sources are pumping current into the top node while the left source is feeding the bottom node. You can also think of the left source as draining the top node. Thus, these three sources together behave like a single source of five amps feeding the top node (eight amps in, three amps out).

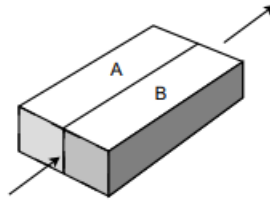
Figure 4.4

Current sources in parallel.



- **Resistors in Parallel:** When placed in parallel, resistor values do not add the way they do in series connections.

Figure 4.5
Resistors in parallel.



Recall that

$$R = \frac{\rho l}{A}.$$

Notice that two of the same resistor in parallel would double the area, which resistivity and length unchanged. Thus, the resistance is halved. So, we see that placing resistors in parallel results in a decrease in net resistance, the opposite of the series case.

Recall that **conductance** G is given by the reciprocal of resistance

$$G = \frac{A}{\rho l}.$$

Now we can see that the increase in area creates a proportional increase in conductivity. If we generalize this for N resistors we find that the equivalent conductance of a group of parallel resistors is their sum

$$G_{\text{total}} = G_1 + \cdots + G_N.$$

Thus,

$$\frac{1}{R_{\text{parallel}}} = \frac{1}{R_1} + \cdots + \frac{1}{R_N}.$$

Hence,

$$R_{\text{parallel}} = \frac{1}{\frac{1}{R_1} + \cdots + \frac{1}{R_N}}.$$

For N resistors of equal resistance,

$$R_{\text{parallel}} = \frac{1}{\frac{N}{R}} = \frac{R}{N}.$$

Also, consider two parallel resistors R_1 and R_2 , with $R_1 < R_2$. Then,

$$R_P = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}.$$

Consider R_1 alone,

$$R_P = \frac{1}{\frac{1}{R_1}} = R_1.$$

Now, consider what happens when we add R_2 . Since resistance is non-negative, adding $1/R_2$ admits

$$\frac{1}{R_1} + \frac{1}{R_2} > \frac{1}{R_1}.$$

Thus,

$$\frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} < \frac{1}{\frac{1}{R_1}} = R_1.$$

Thus, the combined resistance is smaller than the smallest resistor in the parallel series.

- **Product-Sum Rule:** For two resistors,

$$R_P = \frac{R_1 R_2}{R_1 + R_2}.$$

It is only applicable to two resistors, however, it can be used repeatedly on pairs of resistors within larger groups instead of using

Now, suppose we write

$$N = \frac{R_2}{R_1},$$

so that

$$R_2 = \frac{N}{R_1}.$$

Now,

$$R_P = \frac{R_1 N R_1}{R_1 + N R_1} = \frac{N}{N + 1} R_1.$$

This is useful for quick estimates of parallel resistor pairs. If we have a resistor ratio of 3:1, then the equivalent would be 3/4ths the value of the smaller resistor. For a $24k\Omega$ and a $8k\Omega$, the equivalent would be a $6k\Omega$ resistor.

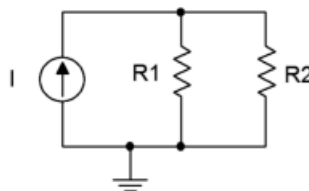
- **Kirchhoff's Current Law (KCL):** Just as Kirchhoff's voltage law is a key element in understanding series circuits, Kirchhoff's current law (KCL) is the operative rule for parallel circuits. It states that the sum of all currents entering and exiting a node must sum to zero. Alternately, it can be stated as the sum of currents entering a node must equal the sum of currents exiting that node. As a pseudo formula:

$$\sum I \leftarrow = \sum I \rightarrow .$$

Recall that a node is a connection area wherein the voltage is the same

Figure 4.7

A simple parallel circuit.



We see that there are two nodes in a parallel configuration. For each of these nodes, whatever currents flow in must be balanced by the same total current flowing out.

There is a current source feeding the top node. At the three-way connection with R_1 and R_2 , KCL along with Ohm's law dictate that the incoming current from the source must split with one portion exiting by flowing down through R_1 and the remainder exiting by flowing down through R_2 .

At the bottom node (ground) these two resistor currents must combine and equal the current that is being drawn back into the current source, which must be the same as the current originally produced by the source. This is perfectly sensible because it would be impossible for the current source to establish two different currents; one at its input and the other at its output. To assert otherwise would be a little like saying you're two different heights depending on whether you measure from your toes up to your head versus from your head down to your toes. Thus, a simplistic way of stating KCL is "What goes in must come out"

- **The Current Divider Rule (CDR):** Just as series circuits follow the voltage divider rule (voltage dividing in proportion to resistance), parallel circuits follow the current divider rule which states that current divides in reverse proportion to resistance (i.e., in direct proportion to conductance).

First, we note that the voltage across the pair must equal the entering current times the effective resistance of the pair.

$$V = I_T \frac{R_1 R_2}{R_1 + R_2}.$$

The current through each of the resistors must be this voltage divided by the corresponding resistance. For the current through R_1

$$I_{R_1} = \frac{V}{R_1} = I_T \frac{R_2}{R_1 + R_2}.$$

Similarly, for the current through R_2 :

$$I_{R_2} = I_T \frac{R_1}{R_1 + R_2}.$$

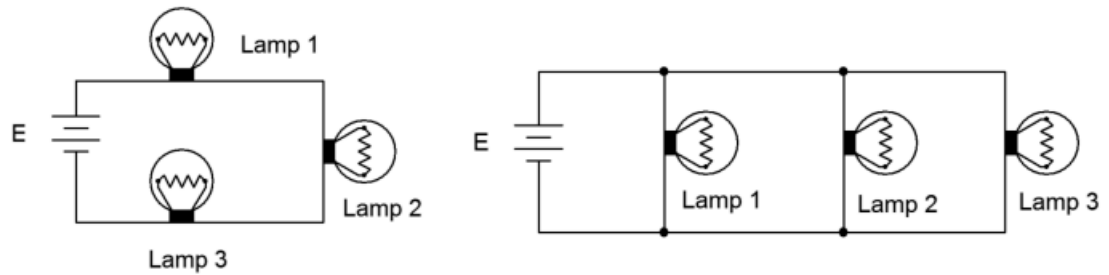
Hence,

$$I_X = I_T \frac{R_Y}{R_X + R_Y}.$$

This rule is convenient in that you don't have to compute the parallel equivalent resistance, but remember, it is valid only when there are just two resistors involved.

- **Fuses and circuit breakers:** Most electrical and electronic devices are designed to run off of a constant voltage source. As such, parallel connections are ideal when a variable number of devices are to be powered because, unlike a series connection, each device will see the same voltage rather than having that voltage split among them.

There is another advantage to parallel connection. Consider



Typically, when a lamp fails, it fails creating an open. If any of the lamps in the series connection fails, current in the loop is interrupted and all of the lamps go out. In contrast, if one of the lamps in the parallel connection fails, the remaining lights will stay on. Further, in the series connection, there is no immediate way to determine which lamp has failed as none of them will be lit. Each lamp has to be tested in turn to determine the faulty unit. In the parallel system, it is obvious which lamp has failed because that's the one that's out.

In a home, school or commercial setting, any number of devices may be plugged into wall sockets with the expectation that each unit will receive the same voltage and operate correctly due to the parallel nature of the system. As convenient as this is, there is a practical problem. Every device that is added to the system presents another path for current flow. Due to KCL, this means that the total current drawn from the source must increase. This creates a potential problem because if too much current is drawn, there can be an appreciable voltage drop across the wires feeding the outlets as they can no longer be assumed to be zero ohms. This lowered voltage may present a problem for the attached devices, but more importantly, the large current results in unexpected heating of the wires because of the power dissipation in them (I^2R) and creates a possible fire hazard. Consequently, we must include some method to limit the current to a safe maximum value.

Adding another load in a parallel circuit increases the total current drawn from the source because it creates a new, separate path for electricity to flow. Each load you add creates an independent branch connected across the same voltage source.

The power source provides the same push (voltage) to every branch. Each new load draws its own separate share of current based on its own resistance

According to Kirchhoff's Current Law, the total current leaving the source is the sum of the currents going through each individual branch. When you add another branch current to the total, the overall sum goes up

Because there are more pathways for the charge to move through, the overall or equivalent resistance of the entire circuit actually decreases.

Since the resistance decreases,

$$I = \frac{V}{R}$$

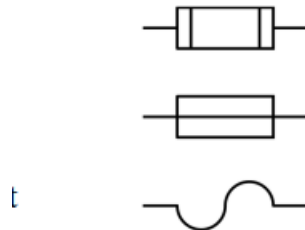
increases. The common approach is a fuse or circuit breaker. These devices are placed between the voltage source and the various loads, and see their combined current. If this current becomes too great, the device interrupts the current flow by opening the circuit. Nothing will operate but nothing catches fire, either.

Fuses are used typically in electronic systems, motors, automotive subsystems and the like. Whether they're the cylindrical type found in electronic instrumentation or the blade type common in automobiles, fuses are one-shot devices. They consist of a fine wire or metal link that will heat up and melt if it experiences too high of a current. Upon melting, circuit continuity is lost and current flow stops. This is called a "blown" fuse. The fuse will then need to be replaced once the problem has been fixed (i.e., whatever it was that caused excessive current to flow in the first place).

Fuses have a nominal current rating, for example, two amps. This does not mean that as soon as two amps is reached, the fuse blows. It will take some time for the fuse to heat up. The higher the current is above the nominal rating, the faster the fuse will blow.



Figure 4.21
Schematic symbols for fuses:
IEC (top and middle), ANSI
(bottom).



Fuses are also available in fast-blow and slow-blow forms. Fast-blow fuses will activate many times faster than standard fuses while slow-blow types will require much longer to activate (this prevents the fuse from activating accidentally with certain kinds of loads that exhibit a high initial startup current but that falls back to a more modest level once running smoothly, such as a motor).

Fuses are generally inexpensive and effective. The obvious downside to a fuse is that they need to be replaced once they do their job. A circuit breaker solves this problem.

A circuit breaker, or just breaker for short, can be thought of as a resettable fuse. Breakers do not have a melting link that interrupts current flow. Instead, a breaker operates like an intelligent switch: if the current is too large, the switch is thrown open. The operator can then fix the problem and reset the breaker (i.e., return the switch to the operating position).

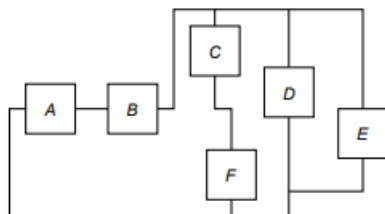
Figure 4.22
*Residential circuit breakers
including cutaway view.*



Series-Parallel Resistive Circuits

- **Intro:** The key to analyzing basic series-parallel circuits is in recognizing portions of the circuit, or sub-circuits, that exhibit a series or parallel configuration by themselves, and then applying the series and parallel analysis rules to those sections. Ohm's law, KVL and KCL may be used in turn to “chip away” at the problem until all currents and voltages are found. As individual voltages and currents are determined, this makes it easier to apply these rules to determine other values. Given this observation, the number of potential solution paths tends to grow exponentially as the number of components increases. As a consequence, when faced with the same circuit, six people may solve it six different ways, no particular way being more or less correct than any other. The only thing we can say is that some solution paths might be more computationally efficient than others, meaning they take less work
- **Series-Parallel Connections:**

Figure 5.1
A series-parallel
configuration.

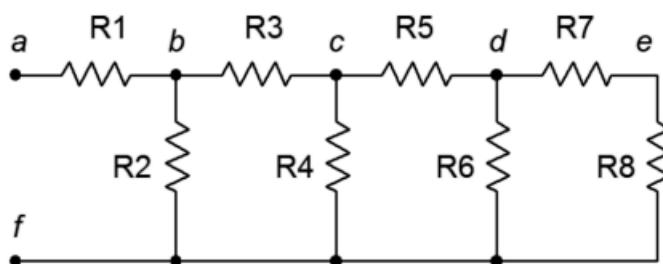


It should be obvious that this system is neither a simple series connection nor a simple parallel connection, but rather some combination of the two. In order for it to be series, the current through each component would have to be the same. This does not have to be true here. Consider the connection node directly above block C. This also connects to items B, D and E. All we really know about that node is that there are four paths leading to it and that the four associated currents must sum to zero, thanks to KCL. There is nothing that requires the current through, say, block D to be the same as that through block E. Similarly, there is nothing that stipulates that the voltage across block C must be the same as that across block F. After all, that is the hallmark of a parallel configuration, and that would require those two blocks to be connected to the same two nodes, which they aren't.

Notice some of the components are in series among themselves, and some are in parallel among themselves. We might recognize these groupings as sub-circuits.

For example, it is true that blocks A and B are in series with each other because the current through one of them must be the same as the current through the other. The same is true of blocks C and F; they are in series with each other. Similarly, we can say that blocks D and E are in parallel with each other because they must see the same voltage as they are each connected to the same two nodes

- **Ladders:** A ladder is a unique series-parallel configuration. It is arranged in a cascade of series and parallel connections.



The main thing to note is that the ladder network consists of “sections loading sections” repeatedly. By loading, we mean that this element (the load) draws off current and power from the prior section. As a result, simple two-resistor voltage dividers cannot be used.

The only proper voltage divider in this configuration is between R_7 and R_8 . The reason for this becomes apparent if we imagine determining the equivalent resistance of the network by placing an ohmmeter at terminals a and f . The ohmmeter will apply a current to node a which passes through R_1 . At node b this current splits, part going down R_2 and part continuing through R_3 . Clearly then, R_1 and R_2 are not in series as they do not see the same current. Further, the current that flows through R_3 enters node c where it splits again, part down R_4 with the remainder flowing through R_5 . Thus, R_3 and R_4 are not in series, either. The same sort of thing happens again at node d . This would continue for as many rungs as are on the ladder, the sole exception being the final two resistors which are in series (R_7 and R_8 here).

How then do we find the equivalent resistance of this network? We begin at the end farthest from the open terminals. We have already noted that R_7 and R_8 are in series. This pair, if treated as a single resistance, is in parallel with R_6 , or

$$R_6 \parallel (R_7 + R_8).$$

This group of three is in series with R_5 , yielding

$$R_5 + (R_6 \parallel (R_7 + R_8)).$$

This new group of four is in parallel with R_4 , yielding

$$R_4 \parallel (R_5 + (R_6 \parallel (R_7 + R_8))).$$

This group is in series with R_3 , and that new group is in turn in parallel with R_2 . Finally, that penultimate grouping is in series with R_1 . Therefore, the equivalent resistance of the network is

$$R_{eq} = R_1 + [R_2 \parallel (R_3 + [R_4 \parallel (R_5 + [R_6 \parallel (R_7 + R_8)])])].$$

Surely, it is a tad tedious to compute the value, but it is not particularly difficult. A word to the wise: if the goal is to determine the voltages at the various nodes or the myriad branch currents, it will be helpful to keep track of the equivalent resistances of the various groupings.

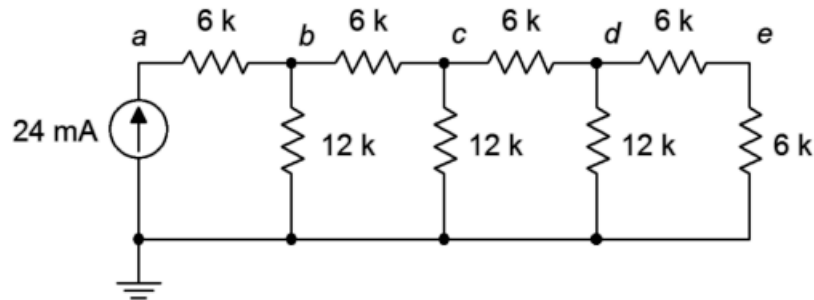
For example, if V_{cf} is known and the goal is to find V_{df} , the voltage-divider relationship would be

$$V_{df} = V_{cf} \frac{R_6 \parallel (R_7 + R_8)}{R_5 + (R_6 \parallel (R_7 + R_8))}.$$

Note that $\parallel$ has higher precedence than $+$ or $-$, therefore $\parallel$ is performed first, rather like multiplication when mixed with addition and subtraction.

- **Any Route Works:** To find some voltage V_{XY} , start at point X and proceed along any convenient route to Y , subtracting voltage rises (polarities of $-$ to $+$) and adding voltage drops (polarities of $+$ to $-$) along the way.
- **R-2R Ladder:** As promised earlier in the chapter, it's time to look at a ladder network. While ladders can use any resistance values desired, certain ratios turn out to be eminently practical. Of particular note is the R-2R ladder

In this configuration, only two different resistor values are used; some initial value and a second value of precisely twice that size



It turns out that this arrangement offers something very unique: a halving of successive node voltages and branch currents. This is extremely useful in digital to analog conversion circuits; the sort of hardware that would turn the digital bits of an MP3 or WAV file into listenable analog signals to feed loudspeakers or headphones.

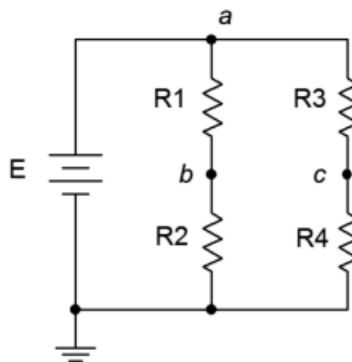
Starting at the far right end, it should be obvious that the last two $6\text{ k}\Omega$ resistors comprise a 50% voltage divider, and thus V_e will half of V_d

The combination of the final three resistors works out to $12\text{ k}\Omega \parallel (6\text{ k}\Omega + 6\text{ k}\Omega)$, or $6\text{ k}\Omega$ again. Thus, the divider from c to d is also 50%. This process repeats as we move to the left, each equivalent working out to R and resulting in a 50% voltage divider. It will not matter what value is chosen for R . As long as $2R$ is used for the other set of resistors, the result will always be successive 50% voltage dividers.

- **Bridges:** A bridge is a particular series-parallel configuration consisting of two pairs of series connected elements placed in parallel.

Figure 5.21

A resistive bridge network.



A typical use for a bridge is the measurement of some environmental quantity. There are certain resistive devices that are sensitive to environmental change. Examples include photoresistors, whose resistance is a function of light level; and thermistors, whose resistance is a function of temperature.

To understand the operation, the voltages V_b and V_c are established by the voltage dividers comprised of R_1 and R_2 ; and R_3 and R_4 , respectively. With ordinary resistors, these voltages would be unchanging and therefore V_{bc} would also be a fixed value. We shall set it up such that V_b and V_c are the same voltage, and thus V_{bc} will be zero. Now assume that we replace R_1 with a photoresistor. As the light level increases, its resistance decreases. This will cause V_b to rise which will cause V_{bc} to go positive. If the light level decreases, the resistance of R_1 will increase, forcing V_b to drop which will cause V_{bc} to go negative. The greater the change in light, the larger V_{bc} will be, and its sign indicates whether the light level has increased or decreased compared to the original set point. By simply placing some manner of voltmeter between points b and c , we can create a display that indicates light levels.

If we want to flip the sign, we would place the photoresistor in the position of R_2 instead of R_1 . Further, it is possible to increase the sensitivity by using sensors at opposite corners (e.g., R_1 and R_4), and comparative or differential measurements are possible by using both sides (e.g., R_1 with R_3).

Analysis theorems and techniques

- **Intro:** We begin by considering a more realistic model for constant voltage and constant current sources. The ideal voltage source produces a stated potential forever, and without regard to what it is connected to. The ideal current source behaves similarly; it will always produce the same current regardless of its load. These expectations are not realistic. For example, if we were to place a solid bar of copper across the terminals of a voltage source, the bar may exhibit a resistance of mere milliohms, implying an output current of thousands of amps. Similarly, if we were to disconnect a current source from any load, its effective load would be the resistance of the air between its terminals and Ohm's law would dictate an output voltage of perhaps thousands or even millions of volts. Real world sources do not behave like this.

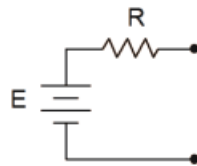
Thus, ideal voltage and current sources cannot exist in the real world because they would theoretically produce infinite or destructive amounts of power.

- **Internal resistance:** A highly precise model for any voltage source or current source could be fairly complex but for general purpose work, we can greatly improve our ideal sources by simply adding a resistor to them. This resistance is referred to as the internal resistance of the source.

It is important to understand that this is not a resistor, as in an internal component that can be altered, but rather a mathematical addition to the source that better predicts how it will behave. Further, the value of this effective resistance can be deduced in a laboratory via proper measurements.

The model for a voltage source adds a resistance in series

Figure 6.1
Practical voltage source model.



This resistance sets an upper limit on the source's current output. Even if the output terminals are shorted, the maximum current will be dictated via Ohm's law to be the source voltage divided by the internal resistance, or E/R . Obviously, this internal resistance will create some voltage divider effect with the attached load. To minimize this effect, the internal resistance should be as small as practicably possible. Thus,

"The ideal internal resistance of a voltage source is zero ohms (a short)."

Given a zero ohm internal resistance, this improved model reverts back to the original ideal source. In the case of a lab power supply, the internal resistance typically will be a small fraction of an ohm.